Start Programming on Raspberry Pi with Python

One of the most revolutionary things that happened in computing in recent times has been the invention of the Raspberry Pi, as it has brought the computer within everyone's reach. As a fallout, there has been a coding revolution. This article is a primer for coding on a Raspberry Pi.

Raspberry Pi is a low-cost computing platform. The goal of the Raspberry Pi Foundation is to make computing available to everyone globally to help them to learn programming. Since its initial release in 2012, the Raspberry Pi has seen several enhancements in terms of the amount of RAM, CPU power, peripheral support, and support for networking protocols; yet, it has managed to hold on to its original US$ 35 price tag (about ₹2400).

The latest version, Raspberry Pi 3, was announced in February 2016. It comes with a 1.2GHz 64-bit quad-core ARMv8 CPU, 1GB RAM, built-in wireless/Bluetooth support and much more. This amount of computing power is more than sufficient to run your applications and to program them using a variety of programming tools/environments. In this article, let's get started with programming on the Raspberry Pi using one of the most popular languages in the world, Python.

Why Raspberry Pi and Python?

The Raspberry Pi has been nothing short of a revolution in introducing millions of people across the world to computing and being one of the drivers behind introducing computer programming to everyone. It has powerful enough hardware to get started with programming and the US$ 35 price tag is hard to beat.

The makers of Raspberry Pi have also paid special attention to ensuring that barriers to getting started are minimal. The recommended Linux distribution for Raspberry Pi, Raspbian, comes bundled with multiple programming languages and IDEs so that you are ready to go from the time you power on the mini development board.

Python, on the other hand, is one of the most popular languages in the world and has been around for more than two decades. It is heavily used in academic environments and is a widely supported platform in modern applications, especially utilities, and desktop and Web applications. Python is highly recommended as a language that is easy for newcomers to program. With its easy-to-read syntax, the introduction is gentle and the overall experience much better for a newbie.

The latest version of the Raspbian OS comes bundled with both Python 3.3 and Python 2.x tools. Python 3.x is the latest version of the Python language and is recommended by the Raspberry Pi Foundation too.

What you can do with Raspberry Pi and Python

The combination of Raspberry Pi and Python can be used for multiple purposes. Some of the popular items include:

- Learning how to program with Python
- Connecting your Raspberry Pi to multiple sensors and receiving data from them or control hardware—for example, home automation, environment monitoring via temperature sensors, etc.
- Using your Raspberry Pi as a Web server with the program written in Python
- Writing various utilities in Python and using your Pi as a server for monitoring and tracking multiple applications, services, etc.

These are just some of the things that you can do. You have the full power to convert your ideas to reality using Raspberry Pi along with Python.

The Python IDLE shell and command line

You can use Python from both an IDE (Integrated Development Environment) and from the terminal, depending on your comfort level. Python comes with a simple IDE called IDLE. Both the Python 2 and 3 IDLE are available, and since this article is focused on Python 3.x, we will look specifically at Python 3 IDLE.

Figure 2 shows that Python 3 IDLE is available and you can launch the IDE by clicking on it.

This will bring up the Python shell as shown in Figure 3. The shell is an REPL (Read Eval Print Loop) that allows you to interactively try out the Python language, including taking inputs from the user.

In the Python shell (Figure 3), you can see multiple Python statements that have been tried out, including the print method, assigning variables and using the addition operator (+).

On the other hand, if you wish to use the Python shell from the terminal, you can do so by opening up a new terminal and then typing 'python3', as shown in Figure 4.

Updating Python packages

Python is known for its strong community and the packages